Electromagnetic Waves

The following subjects will be covered

- EM Wave Propagation in lossless Media (1D).
- EM Wave Propagation in lossy Media (1D).

Back to the EM Wave Equation

$$\vec{\nabla}^2\vec{E} = \mu\sigma\frac{\partial\vec{E}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \qquad \vec{\nabla}^2\vec{H} = \mu\sigma\frac{\partial\vec{H}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}$$

The EM Wave Equation

Consider a source-free $\rho_v = 0$ and lossless (non-conducting) $\sigma = 0$ medium

$$\vec{\nabla}^2\vec{E} = \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \qquad \vec{\nabla}^2\vec{H} = \mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}$$

General expressions of the electric and magnetic field vectors in rectangular coordinates are

$$\vec{E}(x,y,z,t) = E_x(x,y,z,t)\,\vec{a}_x + E_y(x,y,z,t)\,\vec{a}_y + E_z(x,y,z,t)\,\vec{a}_z$$

$$\vec{H}(x,y,z,t) = H_x(x,y,z,t)\,\vec{a}_x + H_y(x,y,z,t)\,\vec{a}_y + H_z(x,y,z,t)\,\vec{a}_z$$

Time-harmonic Wave Equation

Consider the special case where the time variation of the fields/charges/currents is harmonic (sinusoidal).

For generality and convenience, the harmonic behavior is represented in term of exponentials. $e^{j\omega t}$ Sinusoidal, single frequency (ω)

In this case, the fields can be expressed as,

$$\vec{E}(r,t) = \vec{E}_s(r)\, e^{j\omega t}$$

$$\vec{H}(r,t) = \vec{H}_s(r)\, e^{j\omega t}$$

r represent any space coordinate system variables (not necessarily spherical)

Starting with

$$\vec{\nabla}^2\vec{E} = \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2}$$

$$\vec{\nabla}^2\left(\vec{E}_s(r)\, e^{j\omega t}\right) = \mu\varepsilon\frac{\partial^2}{\partial t^2}\left(\vec{E}_s(r)\, e^{j\omega t}\right)$$

$$e^{j\omega t}\,\vec{\nabla}^2\vec{E}_s = -\omega^2\mu\varepsilon\, e^{j\omega t}\,\vec{E}_s$$

$$\vec{\nabla}^2\vec{E}_s = -\omega^2\mu\varepsilon\,\vec{E}_s$$

$$\vec{\nabla}^2\vec{E}_s + \beta^2\,\vec{E}_s = 0$$

Time-harmonic wave equation. A function of space Only.

Helmholtz wave equation

$$\vec{\nabla}^2\vec{H}_s + \beta^2\,\vec{H}_s = 0$$

$$\beta^2 = \omega^2\mu\varepsilon$$

$$\beta = \omega\sqrt{\mu\varepsilon} = \frac{\omega}{v}$$

The Uniform Plane-wave Solution

A general expression of the electric field vector in rectangular coordinates is

$$\vec{E}(x,y,z,t) = E_x(x,y,z,t)\vec{a}_x + E_y(x,y,z,t)\vec{a}_y + E_z(x,y,z,t)\vec{a}_z$$

TH Wave equation $\vec{\nabla}^2\vec{E}(x,y,z) + \beta^2\vec{E}(x,y,z) = 0$ ➡ $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + \beta^2\right)\vec{E}(x,y,z) = 0$

Consider a one-dimensional case (z) with one field component only.

$$\frac{\partial^2 E_x}{\partial z^2} + \beta^2 E_x = 0$$

$$E_x(z) = A\, e^{-j\beta z} + B\, e^{j\beta z}$$

Plane wave Solution

A and B are constants.

Put back the time-harmonic factor

$$E_x(z,t) = A\, e^{j(\omega t - \beta z)} + B\, e^{j(\omega t + \beta z)}$$

Wave travelling along the $+z$ direction

Wave travelling along the $-z$ direction

Take the $+z$ wave

$$E_x(z,t) = E_o\, e^{j(\omega t - \beta z)}$$

$$E_x(z,t) = E_o \cos(\omega t - \beta z) + j\, E_o \sin(\omega t - \beta z)$$

Solution Solution

Take the sine wave solution

$$E_x(z,t) = E_o \sin(\omega t - \beta z)$$

$$\vec{E}(z,t) = E_o \sin(\omega t - \beta z)\ \vec{a}_x$$

Same definitions for E_o, ω, β

Time-harmonic Plane Waves

Electric field Part of the wave

Amplitude

Phase

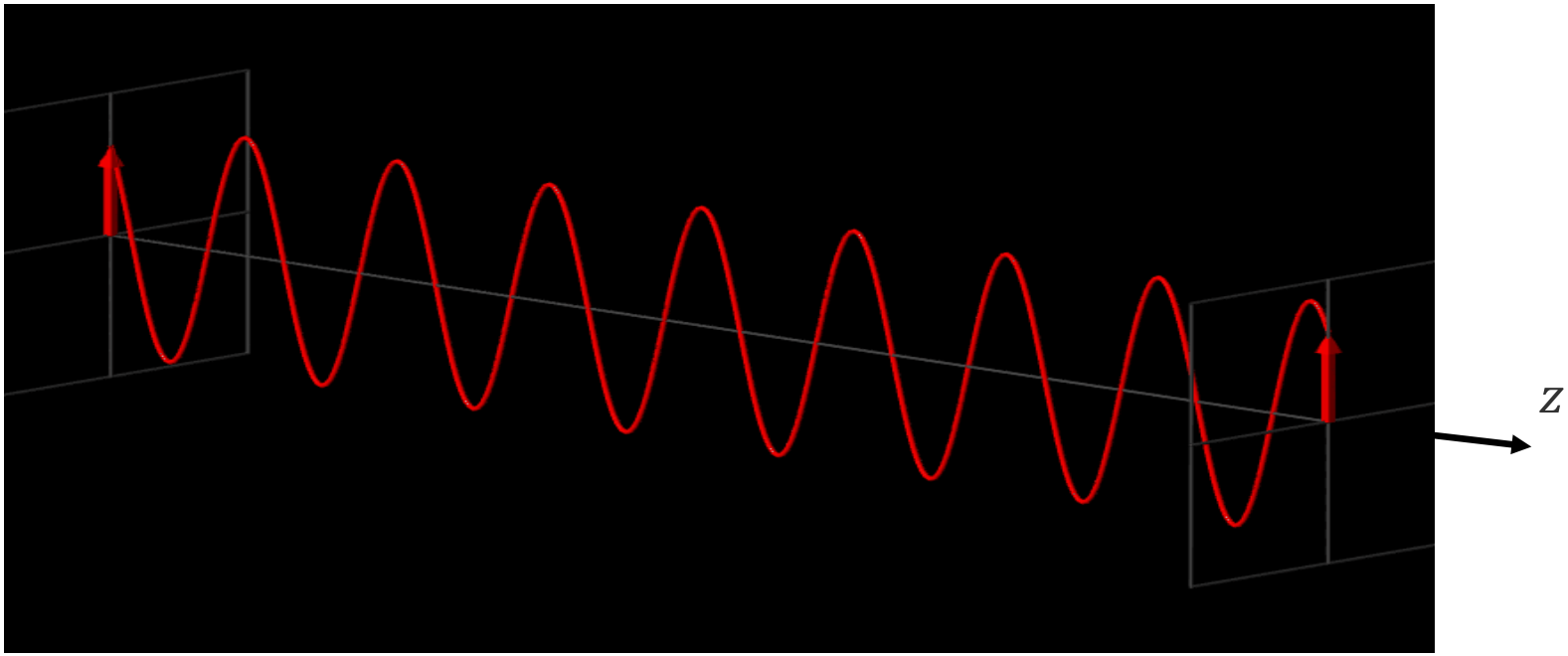
$$\vec{E}(z,t) = E_o \sin(\omega t \pm \beta z)\ \vec{a}_x$$

Field orientation/polarization

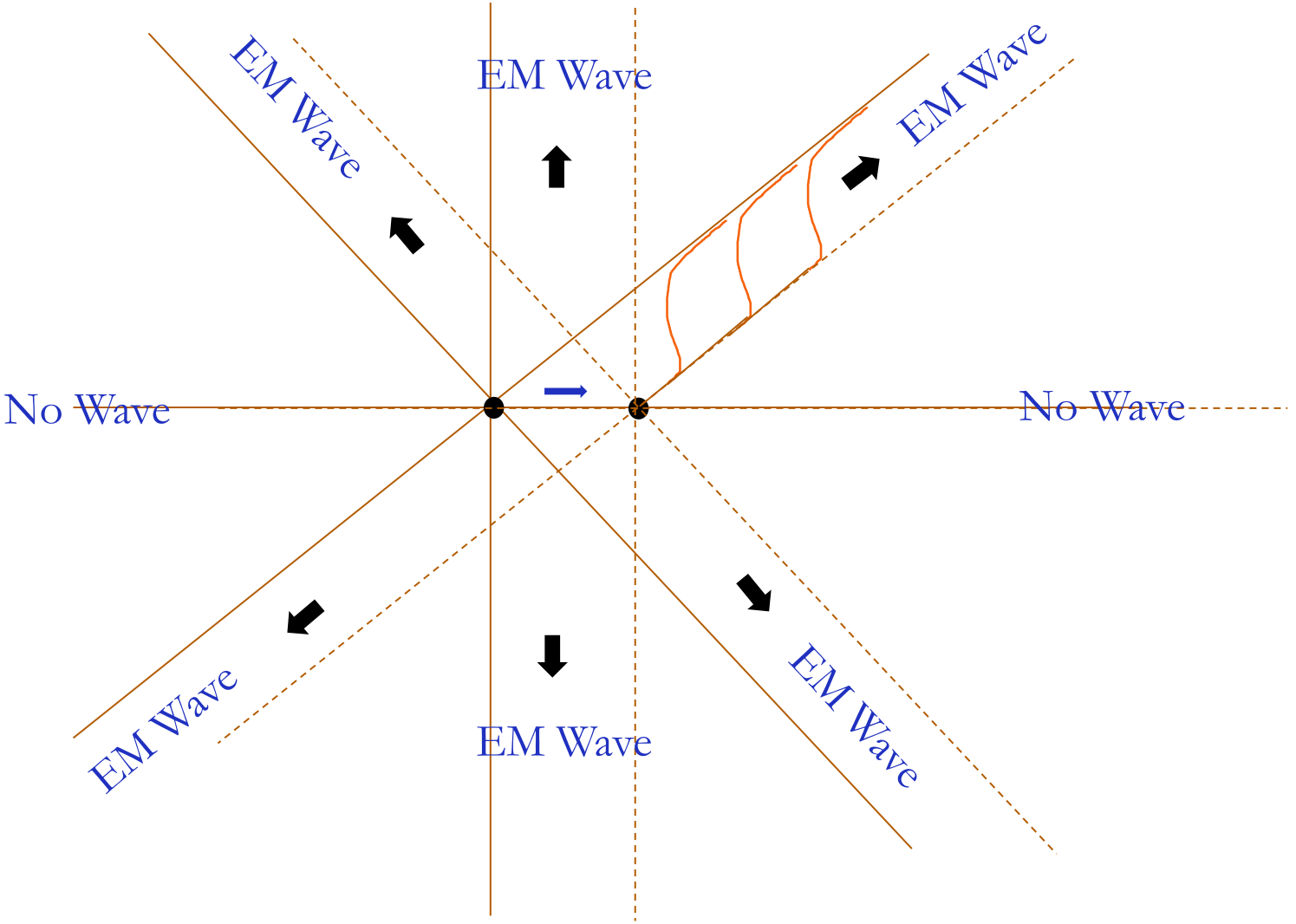
$\omega = 2\pi f$ Frequency

Wave number $\beta = \frac{\omega}{v}$

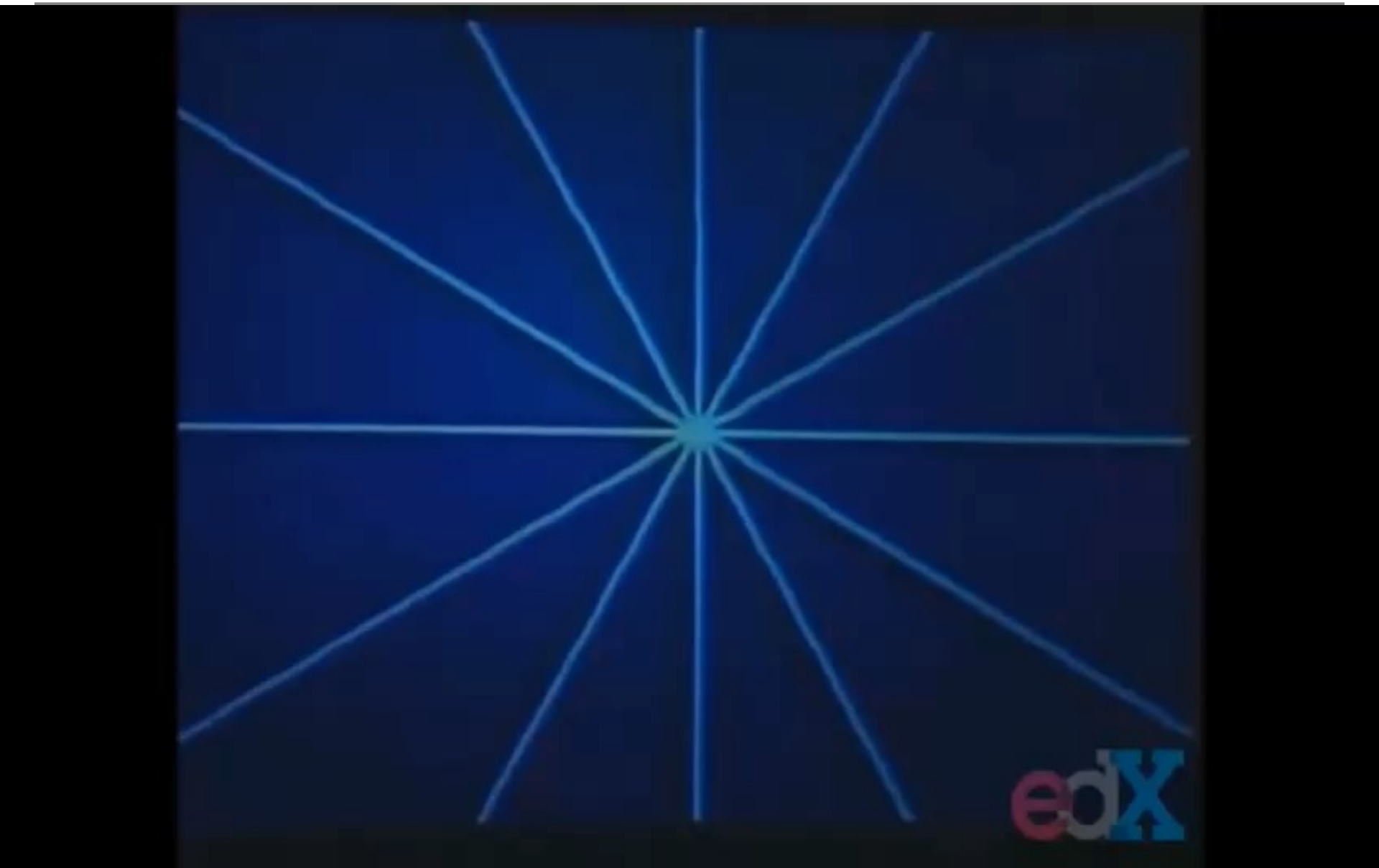
Direction of wave propagation



EM Waves



EM Waves



The Uniform Plane-wave Solution

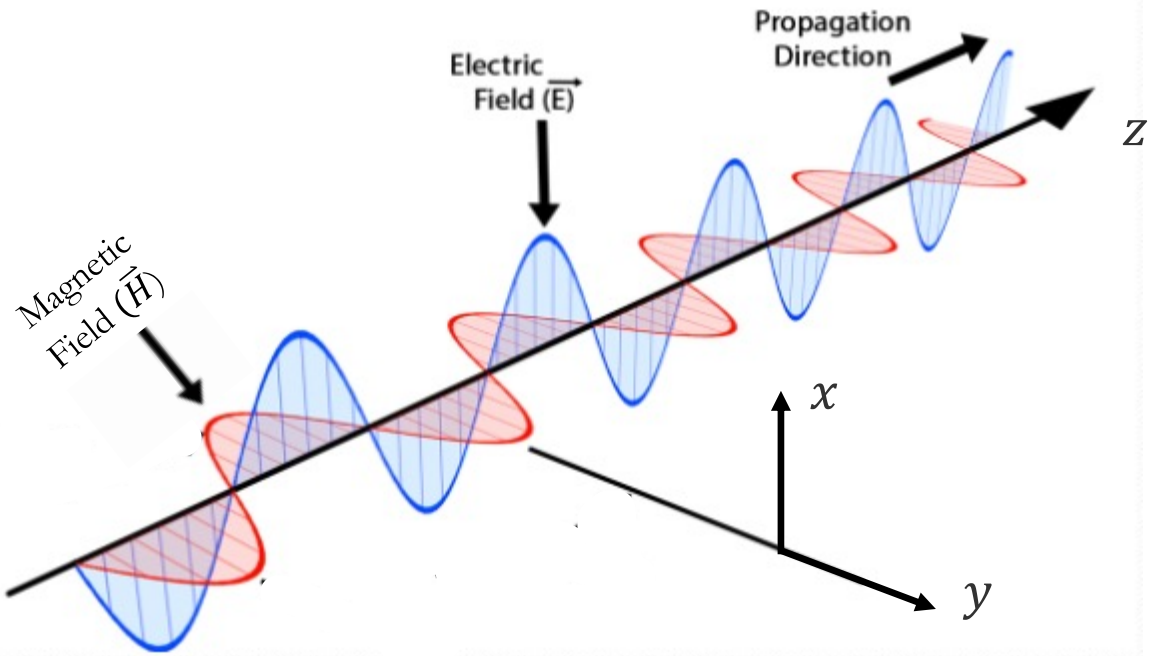
What about $\vec{H}$?

The EM fields are related using Maxwell's equations.

$$\vec{E}(z) = E_o\, e^{-j\beta z}\, \vec{a}_x$$

$$\vec{\nabla} \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t}$$

For the 1D TH case, we have



$$\vec{H} = \frac{1}{-j\omega\mu}\, \vec{\nabla} \times \vec{E} = \frac{1}{-j\omega\mu}\, (-j\beta) E_o\, e^{-j\beta z}\, \vec{a}_y = \frac{\beta}{\omega\mu}\, E_o\, e^{-j\beta z}\, \vec{a}_y \qquad \beta^2 = \omega^2 \mu\varepsilon$$

$$\vec{H} = \sqrt{\frac{\varepsilon}{\mu}}\, E_o\, e^{-j\beta z}\, \vec{a}_y = H_o\, e^{-j\beta z}\, \vec{a}_y$$

$$\eta = \frac{E_o}{H_o} = \sqrt{\frac{\mu}{\varepsilon}} \qquad H_o = \frac{E_o}{\eta} \qquad \vec{a}_k = \vec{a}_E \times \vec{a}_H$$

$\sqrt{\frac{\mu}{\varepsilon}}$ has units of impedance (Ω)

Plane wave impedance

For free-space $\eta_o = \sqrt{\frac{\mu_o}{\varepsilon_o}} = 377\ \Omega$

The Uniform Plane-wave Solution: Summary

For a one-dimensional case (z), the Phasor form solution (for a $+z$ wave) is

$$\vec{E}(z) = E_o\, e^{-j\beta z}\ \vec{a}_E$$

$$\vec{H}(z) = \frac{E_o}{\eta}\, e^{-j\beta z}\ \vec{a}_H$$

$\vec{a}_z = \vec{a}_E \times \vec{a}_H$

For a one-dimensional case (z), the complete form solution is

$$\vec{E}(z,t) = E_o\ e^{j\omega t}\, e^{-j\beta z}\ \vec{a}_E = E_o\, e^{j(\omega t-\beta z)}\ \vec{a}_E$$

$$\vec{H}(z,t) = \frac{E_o}{\eta}\, e^{j\omega t}\, e^{-j\beta z}\ \vec{a}_H = \frac{E_o}{\eta}\, e^{j(\omega t-\beta z)}\ \vec{a}_H$$

$\vec{a}_z = \vec{a}_E \times \vec{a}_H$

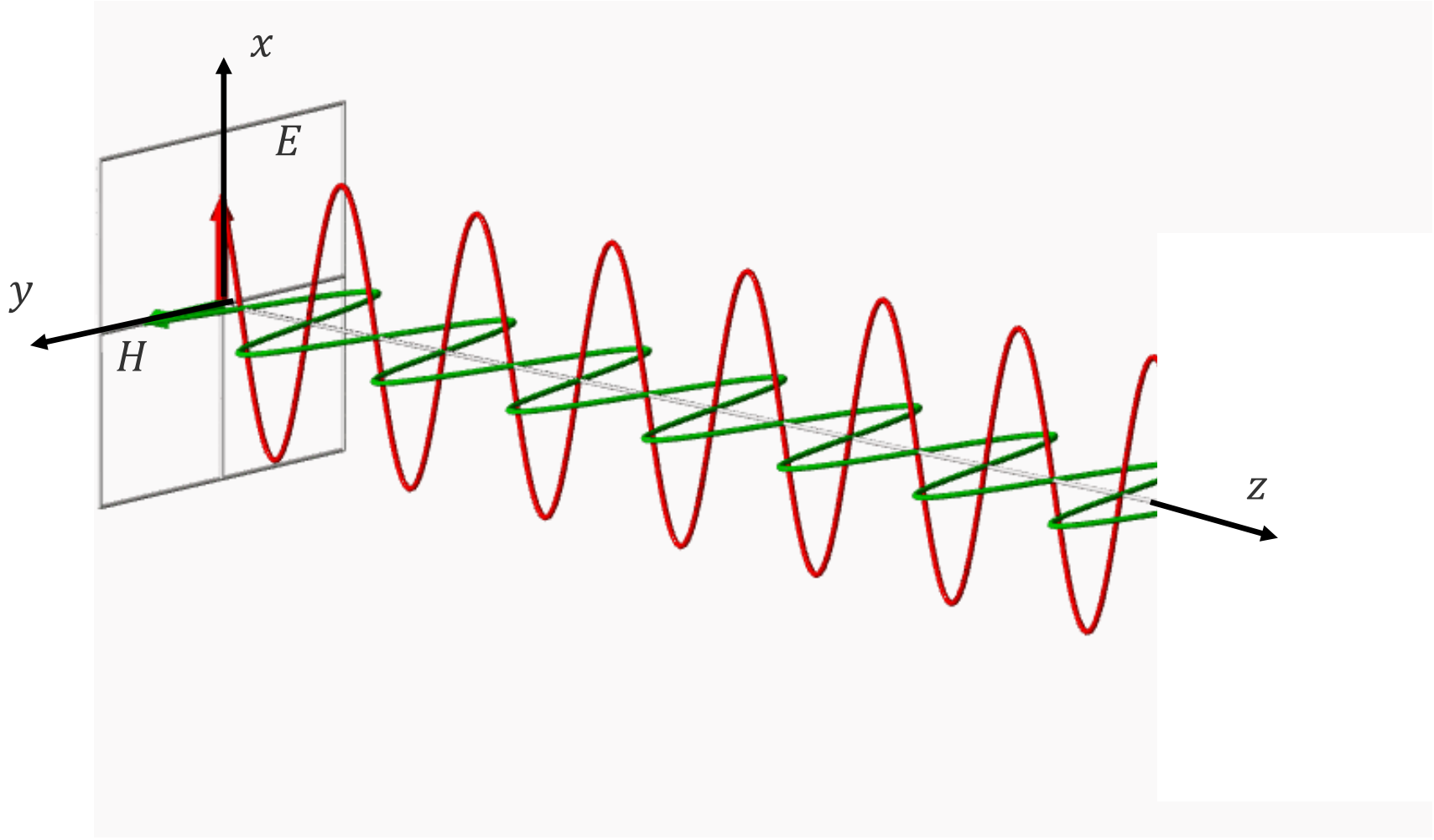
For a one-dimensional case (z), the instantaneous form solution is

$$\vec{\mathcal{E}}(z,t) = Re\{\vec{E}(z,t)\} = E_o \cos(\omega t - \beta z)\ \vec{a}_E$$

$$\vec{\mathcal{H}}(z,t) = Re\{\vec{H}(z,t)\} = \frac{E_o}{\eta}\cos(\omega t - \beta z)\ \vec{a}_H$$

$\vec{a}_z = \vec{a}_E \times \vec{a}_H$

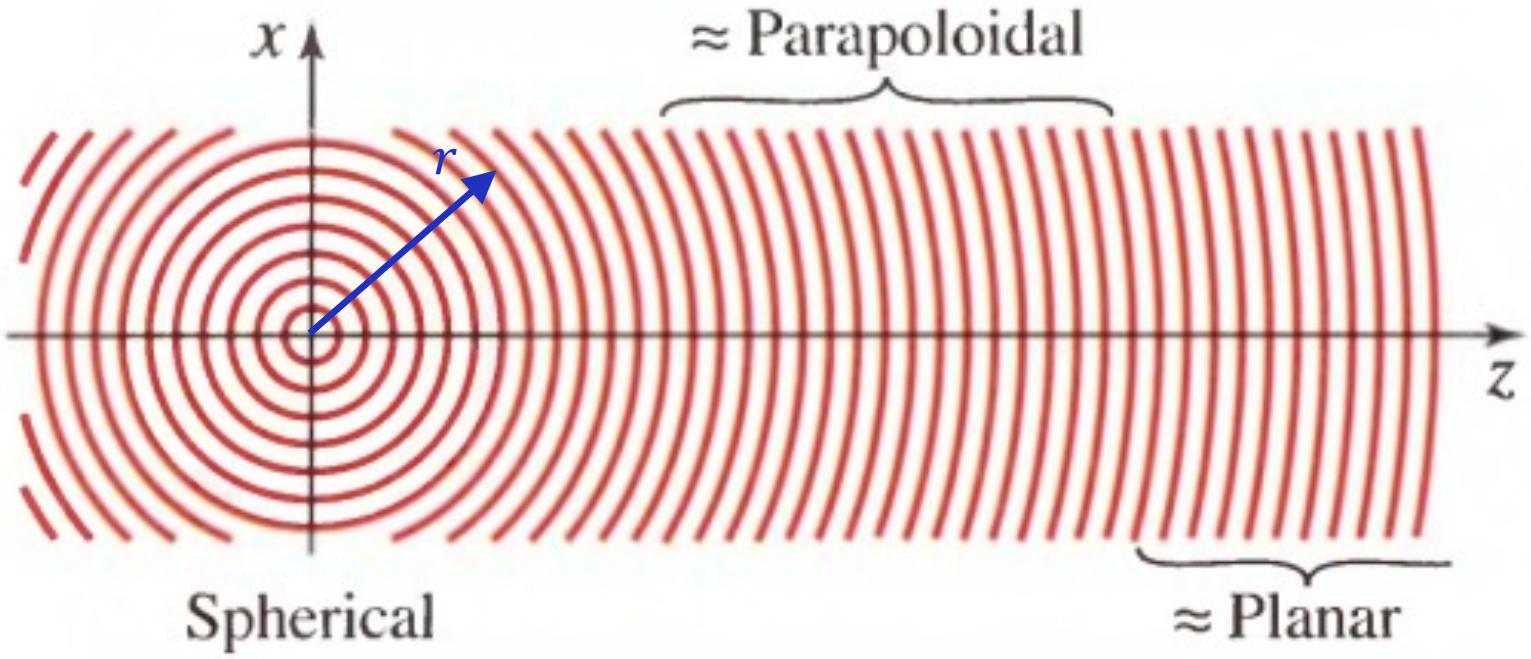
The Uniform Plane-wave Solution



The Wave Equation Solutions

The choice of solution depends on the situation.

$$\vec{E} = \frac{\vec{A}}{z} e^{-j\beta z} e^{-j\beta\left(\frac{x^2+y^2}{2z}\right)}$$

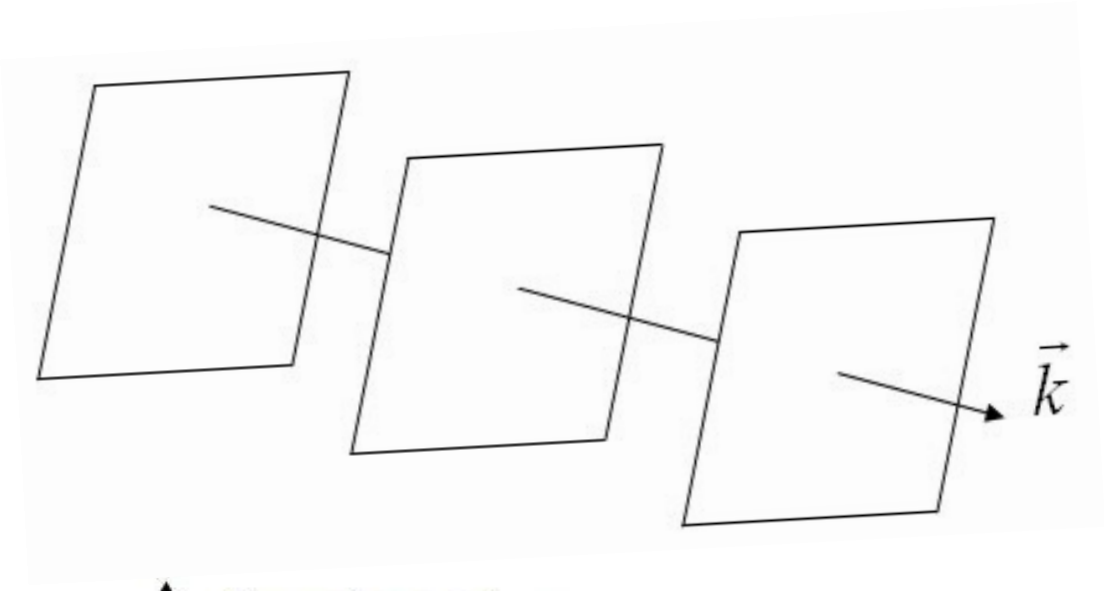


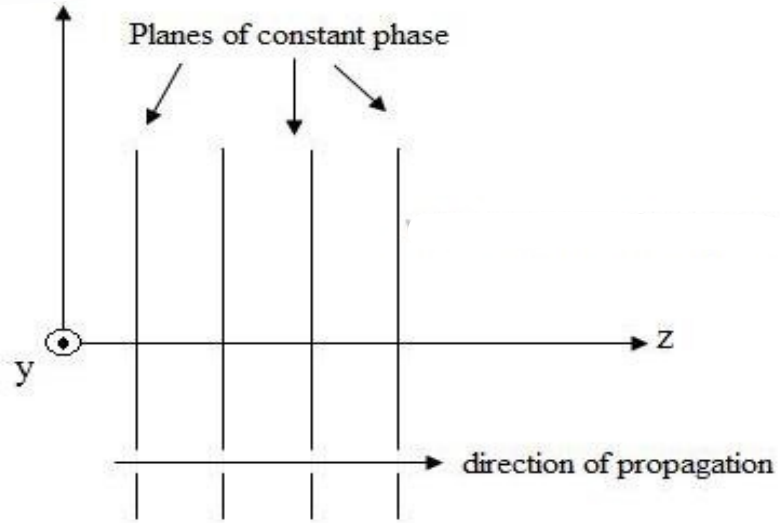
$$\vec{E} = \frac{\vec{A}}{r} e^{-j\beta r}$$

The flashlight example

$$\vec{E} = \vec{A}\, e^{-j\vec{\beta}\cdot\vec{r}}$$

The Uniform Plane-wave Solution





Examples

(1) Consider an EM wave travelling in free space. The wave is polarized in the y-direction and is given by

$$\vec{E}(x,t) = 50\cos(10^8 t + \beta x)\ \vec{a}_y \quad \text{(V/m)}$$

The direction of propagation, $-x$ direction.

Frequency $\omega = 10^8$ rad/s.

Phase constant (wavenumber) $\beta = \frac{\omega}{v} = \frac{10^8}{3\times10^8} = \frac{1}{3}$ rad/m.

How many meters to complete a full 2π phase?

Wavelength $\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{1/3} = 6\pi$ m.

Write the expression for the magnetic field. $H_o = \frac{E_o}{\eta_o} = \frac{50}{377}$

$\vec{a}_k = \vec{a}_E \times \vec{a}_H$

$-\vec{a}_x = \vec{a}_y \times \vec{a}_H$

$$\vec{H}(x,t) = \frac{50}{377}\cos\left(10^8 t + \frac{1}{3}x\right)(-\vec{a}_z) \quad \text{(A/m)}$$

Examples

(2) The electric field intensity of a time-harmonic wave travelling in a source-free glass (μ_r=1, ε_r=5) is given by $\vec{E} = (80\,\vec{a}_x - 60\,\vec{a}_z)\,e^{j8y}$ (V/m)

What is the wave propagation direction? $-y$ direction.

What is the wavelength in the wave propagation direction?

$\beta = 8$ rad/m $\qquad \lambda = \frac{2\pi}{\beta} = 0.785$ m

Find the frequency of the wave. $f = \frac{v}{\lambda} = \frac{c/\sqrt{\varepsilon_r}}{\lambda} = 171\ MHz$

Find the corresponding magnetic field?

$\eta = \sqrt{\frac{\mu}{\varepsilon}} = \frac{\eta_o}{\sqrt{\varepsilon_r}} = \frac{377}{\sqrt{5}} = 75.4\ \Omega$

$\vec{H} = \frac{-1}{75.4}(80\,\vec{a}_z + 60\,\vec{a}_x)\,e^{j8y}$ (A/m)

$\vec{H} = \frac{-1}{75.4}(80\,\vec{a}_z + 60\,\vec{a}_x)\cos(171\times10^6 t - 8y)$ (A/m)

The Plane-wave Solution in Lossy Media

$$\vec{\nabla}^2\vec{E} = \mu\sigma\frac{\partial\vec{E}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2}$$

Consider a source-free $\rho_v = 0$
and lossy (conducting) $\sigma \neq 0$ medium

Time-harmonic waves are generated by time-harmonic sources.

$$\vec{E} = \vec{E}_s\, e^{j\omega t}$$

$$\vec{\nabla}^2\vec{E}_s \cdot e^{j\omega t} = \mu\sigma\vec{E}_s\,(j\omega)\cdot e^{j\omega t} + \mu\varepsilon\vec{E}_s\,(j\omega)^2\cdot e^{j\omega t}$$

$$\vec{\nabla}^2\vec{E}_s = j\omega\mu\sigma\vec{E}_s + \mu\varepsilon\,(j\omega)^2\vec{E}_s = j\omega\mu(\sigma + j\omega\varepsilon)\vec{E}_s$$

$$\vec{\nabla}^2\vec{E}_s = j\omega\mu(\sigma + j\omega\varepsilon)\vec{E}_s$$

$$\vec{\nabla}^2\vec{H}_s = j\omega\mu(\sigma + j\omega\varepsilon)\vec{H}_s$$

$$\gamma^2 = j\omega\mu(\sigma + j\omega\varepsilon)$$

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)} = \alpha + j\beta$$

$$\vec{\nabla}^2\vec{E}_s - \gamma^2\vec{E}_s = 0$$

$$\vec{\nabla}^2\vec{H}_s - \gamma^2\vec{H}_s = 0$$

Helmholtz wave equation
General Media

Propagation Coefficient
Complex quantity

The Plane-wave Solution in Lossy Media

A general expression of the electric field vector in rectangular coordinates is

$$\vec{E}(x,y,z,t) = E_x(x,y,z,t)\vec{a}_x + E_y(x,y,z,t)\vec{a}_y + E_z(x,y,z,t)\vec{a}_z$$

Consider a one-dimensional case (z) with one component only.

$$\frac{\partial^2 E_x}{\partial z^2} - \gamma^2 E_x = 0$$

$$E_x(z) = A\, e^{-\gamma z} + B\, e^{\gamma z}$$

Wave travelling along the $+z$ direction

Wave travelling along the $-z$ direction

Take the $+z$ wave

$$E_x(z) = E_o\, e^{-(\alpha + j\beta)z}$$

$$E_x(z) = E_o\, e^{-\alpha z}\; e^{-j\beta z}$$

Attenuation Phase

Take the sine wave solution

$$E_x(z,t) = E_o \sin(\omega t - \beta z)\, e^{-\alpha z}$$

$$\vec{E}(z,t) = E_o \sin(\omega t - \beta z)\, e^{-\alpha z}\; \vec{a}_x$$

$$\vec{H}(z,t) = H_o \sin(\omega t - \beta z)\, e^{-\alpha z}\; \vec{a}_H$$

The Propagation Coefficient

Starting with the basic definition,

$$\gamma^2 = j\omega\mu(\sigma + j\omega\varepsilon) = -\omega^2\mu\varepsilon + j\omega\mu\sigma$$

complex quantity

$$\gamma = \alpha + j\beta$$

$$\gamma^2 = \alpha^2 + 2j\alpha\beta + \beta^2 = (\alpha^2+\beta^2) + j(2\alpha\beta)$$

Solve for α and β,

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}-1\right]}$$

Attenuation Coefficient Np/m

$$\beta = \omega\sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}+1\right]}$$

Phase Coefficient Rad/m

The Plane-wave Solution in Lossy Media

What about $\vec{H}$?

The EM fields are related using Maxwell's equations.

$$\vec{\nabla} \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t} \quad \rightarrow \quad \vec{H} = \frac{1}{-j\omega\mu} \vec{\nabla} \times \vec{E}$$

$$\vec{E}(z) = E_o \, e^{-\gamma z} \, \vec{a}_x$$

$$\vec{H}(z) = \frac{1}{-j\omega\mu} \frac{d(E_o \, e^{-\gamma z})}{dz} \vec{a}_y$$

$$\vec{H}(z) = \frac{\gamma}{j\omega\mu} E_o \, e^{-\gamma z} \, \vec{a}_y$$

$$\vec{H}(z) = H_o \, e^{-\gamma z} \, \vec{a}_y$$

$$\eta = \frac{E_o}{H_o} = \frac{j\omega\mu}{\gamma} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}}$$

Wave impedance in a lossy medium

$$\eta = |\eta| \angle \theta_\eta$$

$$\tan(2\theta_\eta) = \frac{\sigma}{\omega\varepsilon}$$

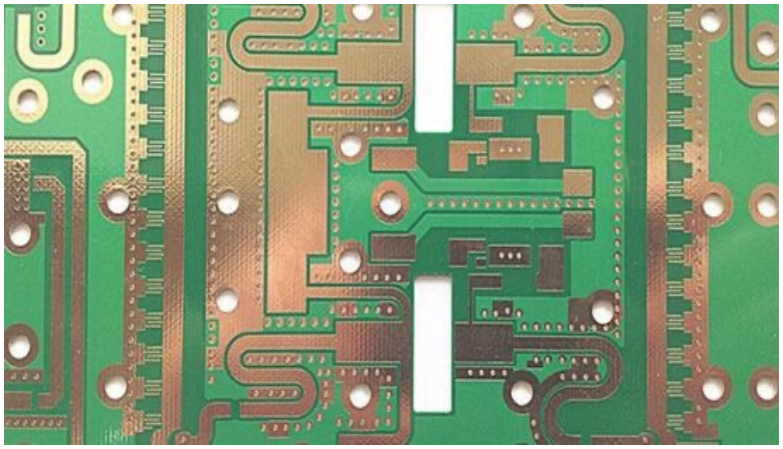
Wave Impedance

For general materials

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = |\eta| \angle\theta_\eta$$

$$|\eta| = \frac{\sqrt{\mu/\varepsilon}}{\left[1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2\right]^{1/4}}$$

$$\tan 2\theta_\eta = \frac{\sigma}{\omega\varepsilon}$$



$$H_o = \frac{E_o}{|\eta| \angle\theta_\eta}$$

A phase may be introduced between E and H due to medium

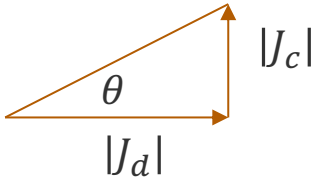
The loss tangent

A measure of how a material is conductive is given by the loss tangent. It is defined as,

$$\tan\theta = \frac{|J_c|}{|J_d|} = \frac{|\sigma E|}{|j\omega\varepsilon E|} = \frac{\sigma}{\omega\varepsilon} = \frac{\varepsilon_r''}{\varepsilon_r'}$$

Loss tangent

$$\theta = 2\theta_\eta$$



Quartz has dielectric constant 3.8, and loss tangent of around 10^{-4} for frequencies from 100 MHz to 10 GHz.

FR4 has dielectric constant 4.4, and loss tangent of 0.030 at 1 GHz.

Classification of Media

Free space $\varepsilon = \varepsilon_o = 8.85\times10^{-12}$ F/m $\mu = \mu_o = 4\pi\times10^{-7}$ H/m $\sigma = 0$

$\eta = \eta_o = 377\ \Omega$ $v = c = 3\times10^{8}$ m/s $\beta = \beta_o = \omega\sqrt{\varepsilon_o\mu_o}$ $\lambda = \lambda_o = 2\pi/\beta_o$

Lossless Medium $\varepsilon = \varepsilon_r\varepsilon_o$ $\mu = \mu_r\,\mu_o$ $\sigma = 0$

$\eta = \eta_o\sqrt{\mu_r/\varepsilon_r}$ $v = c/\sqrt{\varepsilon_r}$ $\beta = \omega\sqrt{\varepsilon_o\varepsilon_r\mu_o}$ $\lambda = 2\pi/\beta$

Lossy/general Medium $\varepsilon = \varepsilon_o(\varepsilon_r' - j\varepsilon_r'')$ $\mu = \mu_r\,\mu_o$ $\sigma \neq 0$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} \qquad \alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1\right]} \qquad \beta = \omega\sqrt{\frac{\mu\varepsilon}{2}\left[\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} + 1\right]}$$

$\gamma = \alpha + j\beta$ $v = \omega/\beta$ $\lambda = 2\pi/\beta$

To classify material, use loss tangent

$\frac{\sigma}{\omega\varepsilon} \ll 1$	$\frac{\sigma}{\omega\varepsilon} \gg 1$	$\frac{\sigma}{\omega\varepsilon} \approx 1$
Lossless	Conductors	Lossy material

Classification of Media

Very good conductors $\varepsilon = \varepsilon_o$ $\mu = \mu_r \mu_o$ $\sigma = \infty$

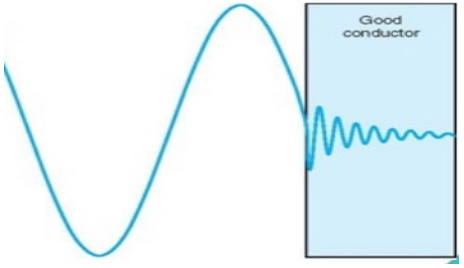
$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = \sqrt{\frac{j\omega\mu}{\sigma}} \qquad \alpha = \sqrt{\frac{\omega\mu\sigma}{2}} \qquad \beta = \sqrt{\frac{\omega\mu\sigma}{2}} \qquad v = \sqrt{\frac{2\omega}{\mu\sigma}}$$

Skin depth A measure of how good is a conductor.

$$\delta = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f \mu \sigma}}$$ Skin depth

$$\eta = \frac{\sqrt{2}}{\sigma\delta} e^{j\pi/4}$$

$$\eta = \frac{1+j}{\sigma\delta}$$



Copper $\sigma = 5.8\times10^7$ S/m

f	10^2	10^8	10^{10}
$\delta\ (mm)$	6.6	$6.6\text{x}10^{-3}$	$6.6\text{x}10^{-4}$

Examples

(1) Consider an EM wave travelling in a lossy dielectric of impedance $200\angle 30°\ \Omega$ with some frequency. At this frequency, the magnetic field of the wave is given by

$$\vec{H}(x,t) = 10\, e^{-\alpha x} \cos(\omega t - x/2)\ \vec{a}_y \quad \text{(A/m)}$$

The direction of propagation is $+x$ direction $\vec{a}_k = \vec{a}_x$.

Expression for E. $\quad E_o = \eta H_o \quad E_o = 200\angle 30°\ (10) = 2000\, e^{j\pi/6}$ V/m

$\beta = 1/2$ rad/m $\quad \vec{a}_k = \vec{a}_E \times \vec{a}_H \quad \vec{a}_x = \vec{a}_E \times \vec{a}_y \rightarrow \vec{a}_E = -\vec{a}_z$

$$\frac{\alpha}{\beta} = \left(\frac{\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1}{\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} + 1} \right)^{1/2} = \frac{1}{\sqrt{3}}$$

Note that $\quad \tan 2\theta_\eta = \frac{\sigma}{\omega\varepsilon} = \sqrt{3}$

$\alpha = \frac{1}{2\sqrt{3}}$ Np/m $\quad \vec{E}(x,t) = 2\, e^{j\pi/6}\, e^{-\frac{x}{2\sqrt{3}}} \cos(\omega t - x/2)\ (-\vec{a}_z)$ kV/m

Q & A

Send me an email if you have any questions.